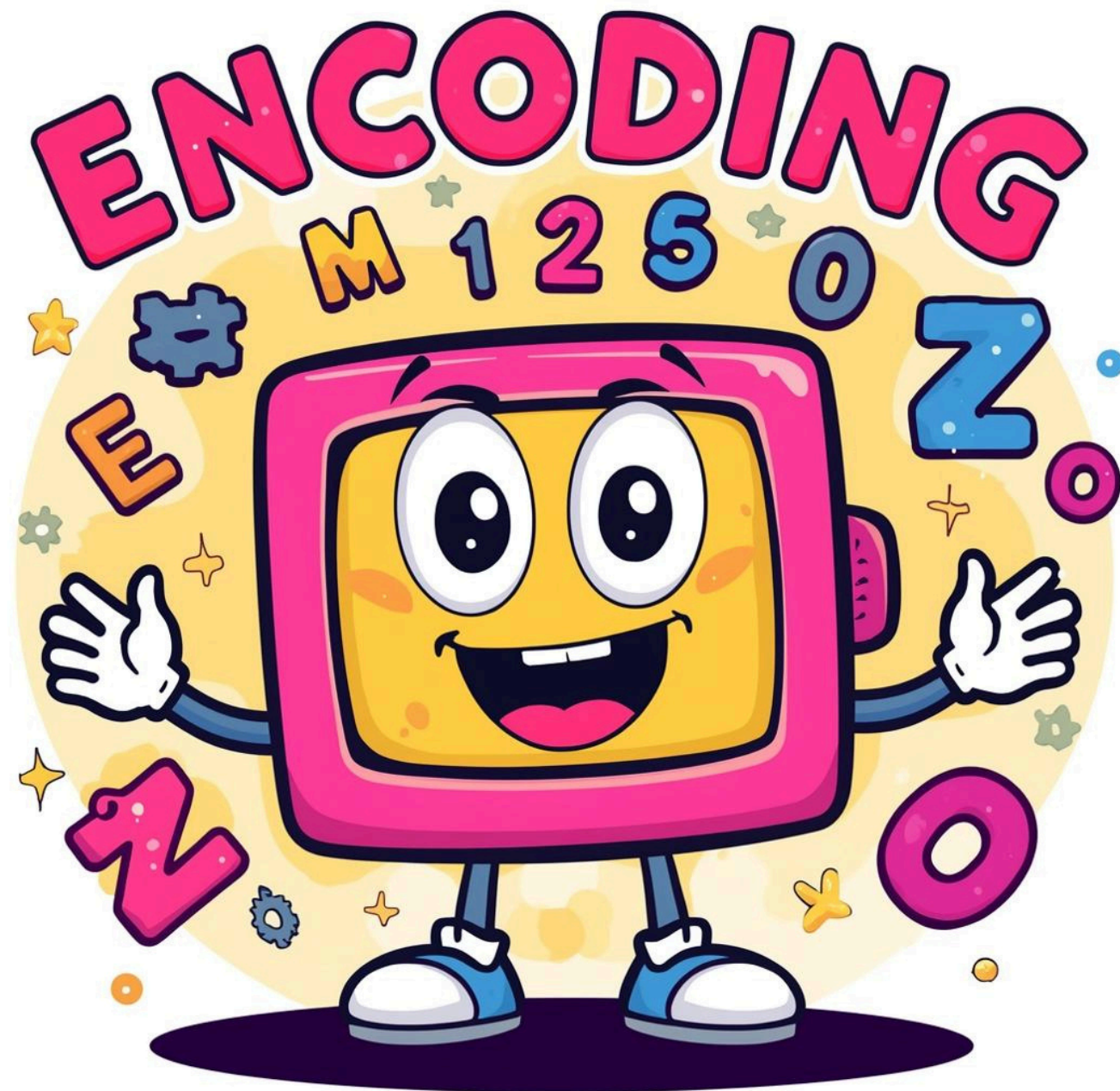


Encoding Explained



Introduction to Encoding

Understanding How Letters and Sounds Become Numbers for Computers

Encoding is like a magic trick that helps computers understand **human language**! It turns letters, pictures, and sounds into numbers, which computers can easily work with. For example, the letter 'A' becomes the number 65. This way, computers can read and process all the cool stuff we create!



Why Computers Need Encoding



Understanding Us

Encoding helps computers make sense of our world!



Character Encoding

ASCII

ASCII turns letters into numbers simply.



UTF-8

UTF-8 can handle many languages and emojis.



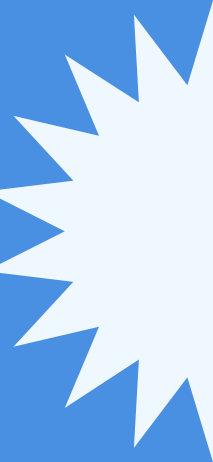
Emojis

Emojis bring fun and emotions into text!



Numerical Encoding

Understanding Categories in Computer Language



Label Encoding

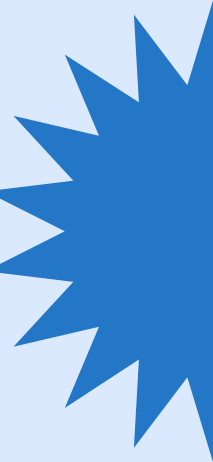
Label encoding helps computers turn categories like colors into single numbers. For example, Red becomes 0, Blue becomes 1, and Green becomes 2. While this makes it easy for computers to handle data, it can create problems because the numbers suggest a fake order. For instance, 0 is not really “less” than 1!

One-Hot Encoding

One-hot encoding improves how computers understand categories by turning them into lists of zeros and ones. For example, Red would be represented as [1, 0, 0]. This method is better for some computer learning tasks since it shows that different categories are unique, but it also uses more space in memory.

Encoding Text for AI

Understanding Words as Numbers for Computers



Bag-of-Words

Bag-of-Words is a way computers count how many times words appear in a text. It makes a list of all the words and the number of times each word shows up. Although it's simple and easy to understand, it doesn't capture the meaning behind the words, making it less effective for understanding context.

Word Embeddings

Word Embeddings take things a step further by turning words into lists of numbers that capture their meanings. For example, if you subtract "Man" from "King" and add "Woman," you get "Queen." This helps computers understand relationships between words, making them smarter when interpreting sentences and ideas.

Why Encoding Matters

Computers Think

Encoding helps computers **understand human language**.



Smarter Machines

Better encoding means **smarter games and robots**.

Summary of Encoding: Making Sense of Numbers



Turning Ideas into Numbers

Encoding helps computers **understand language** and information better.



Explore

Keep Exploring How
Technology Works
Together!